

Formalisation of “Proof” via Monomorphisms, Epimorphisms and Contractions

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Abstract

We show that the act of proving can be described rigorously by

- a monomorphism (injective embedding) $\iota: \mathcal{R} \hookrightarrow \mathcal{P}$,
- an epimorphism (surjective projection) $\phi: \mathcal{P} \twoheadrightarrow \mathcal{R}$,
- and a contraction property of ϕ with respect to the distance $\|X\| = K(X) + \ell(X)$.

Together these give a categorical picture of proof, guarantee loss-free information transfer, and ensure monotone decrease of the distance until the unique 0-dimensional fixed point $\mathbf{o}$ is reached. We use this categorical core as the convergence mechanism behind the *Catuskoti* (tetralemma) interpretation: distinct truth statuses are understood as collapsing, via evaluation, to the same 0-dimensional fixed point (“0d”) $\mathbf{o}$. In particular, we do not treat “emptiness” as a separate object: it is simply this 0-dimensional fixed point.

1 What becomes more rigorous

Definition 1.1 (Objects).

$$\mathcal{R} := \{ (s_x, s_y) \mid s_x, s_y \in \Sigma \} \quad (\text{static, time-free information})$$

$$\mathcal{P} := \{ ((s_x, t_x), (s_y, t_y)) \mid (s_x, s_y) \in \mathcal{R}, t_x, t_y \in \mathcal{T} \} \quad (\text{proof objects, time-axis of length 1}).$$

Here Σ is a set of bit-strings (natural numbers, symbols, ...) and $\mathcal{T}$ is a set of time tokens (any objects with $\ell(t) = 1$).

Definition 1.2 (Embedding and projection).

$$\begin{aligned} \iota: \mathcal{R} &\hookrightarrow \mathcal{P}, & \iota(s_x, s_y) &:= ((s_x, t_x), (s_y, t_y)), \quad t_x, t_y \in \mathcal{T} \text{ arbitrary;} \\ \phi: \mathcal{P} &\twoheadrightarrow \mathcal{R}, & \phi((s_x, t_x), (s_y, t_y)) &:= (s_x, s_y). \end{aligned}$$

Definition 1.3 (Distance (axiomatised)). We fix two maps

$$K: \mathcal{R} \cup \mathcal{P} \rightarrow \mathbb{N}, \quad \ell: \mathcal{R} \cup \mathcal{P} \rightarrow \{0, 1\}$$

and define the distance

$$\|X\| := K(X) + \ell(X).$$

We require the following conventions.

(D1) $\ell(x) = 0$ for all $x \in \mathcal{R}$ and $\ell(p) = 1$ for all $p \in \mathcal{P}$.

(D2) (Time erasure does not change K .) For all $p \in \mathcal{P}$,

$$K(\phi(p)) = K(p).$$

In this note, K is treated as an abstract description length rather than the full Kolmogorov complexity (so the above identity is admissible as a modelling assumption).

Definition 1.4 (Monoid-type properties). • ι is a **monomorphism** (left-cancellable injection); consequently no information is lost: $\iota(x) = \iota(y) \Rightarrow x = y$.

- ϕ is an **epimorphism** (right-cancellable surjection); every static object is the image of some proof.
- ι is a **split monomorphism**: there exists $\sigma : \mathcal{P} \rightarrow \mathcal{R}$ with $\sigma \circ \iota = \text{id}_{\mathcal{R}}$. One can take $\sigma((s_x, t_x), (s_y, t_y)) = (s_x, s_y)$.
- ϕ is a **split epimorphism**: there exists $\rho : \mathcal{R} \rightarrow \mathcal{P}$ with $\phi \circ \rho = \text{id}_{\mathcal{R}}$. Take $\rho = \iota$.
- $\iota \dashv \phi$ (they form an adjoint pair); ι adds a unit of time, ϕ removes it.

Definition 1.5 (Contraction (time erasure)). For every $p \in \mathcal{P}$,

$$\|\phi(p)\| = \|p\| - 1.$$

This is immediate from (D1)–(D2): ϕ removes exactly one unit of “time length” while preserving K .

Definition 1.6 (A discrete evaluation operator on $\mathcal{R}$). To speak about convergence to $\mathbf{o}$ inside the time-free space $\mathcal{R}$, we postulate a map

$$\text{Eval} : \mathcal{R} \rightarrow \mathcal{R}$$

with the following properties:

(E1) (Fixed point) $\text{Eval}(\mathbf{o}) = \mathbf{o}$.

(E2) (Strict decrease away from $\mathbf{o}$) For all $x \in \mathcal{R} \setminus \{\mathbf{o}\}$,

$$\|\text{Eval}(x)\| = \|x\| - 1.$$

This can be seen as a discrete analogue of a contraction step (e.g. “delete one bit of description”), and is the only additional ingredient needed to make the limit claim fully formal.

2 Formal statement that a proof “exists”

Theorem 2.1 (Catuskoti 0d-isomorphism). Let $x \in \mathcal{R}$. Define the sequence in the static space

$$v_0 := x, \quad v_{k+1} := \text{Eval}(v_k) \quad (k \geq 0).$$

Then $\{\|v_k\|\}_{k \geq 0}$ is a strictly decreasing sequence of non-negative integers until it reaches 0, hence there exists $N \in \mathbb{N}$ such that

$$v_N = \mathbf{o}, \quad \text{and therefore } \lim_{k \rightarrow \infty} v_k = \mathbf{o}.$$

Moreover, the categorical maps $\iota : \mathcal{R} \hookrightarrow \mathcal{P}$ and $\phi : \mathcal{P} \rightarrow \mathcal{R}$ provide the time-axis book-keeping: ι adds one unit of time and ϕ removes it, satisfying $\|\phi(p)\| = \|p\| - 1$ for all $p \in \mathcal{P}$.

Proof. By (E2), if $v_k \neq \mathbf{o}$ then $\|v_{k+1}\| = \|v_k\| - 1$; by (E1) the sequence stays at $\mathbf{o}$ once reached. Since $\|v_0\| \in \mathbb{N}$ and decreases by 1 at each step until it hits 0, after $N = \|v_0\|$ steps we obtain $\|v_N\| = 0$. By definition of $\mathbf{o}$ as the unique element of distance 0, we have $v_N = \mathbf{o}$. The statement about ϕ is exactly the contraction property proved from (D1)–(D2). $\square$

3 Summary

- The act of proving is captured categorically by a **monomorphism** ι (information-preserving embedding) and an **epimorphism** ϕ (information-recovering projection).
- Both maps are **split**, guaranteeing exact recovery of the original static data.
- The distance $\|X\| = K(X) + \ell(X)$ provides a quantitative measure of “information + time”. The evaluation map ϕ is a **contraction**, ensuring monotone decrease of this distance.
- Iterating $\phi \circ \iota$ yields a strictly decreasing integer sequence that inevitably reaches the unique 0-dimensional fixed point $\mathbf{o}$.
- Hence the statement “a proof exists for (s_x, s_y) ” is precisely the existence of the morphisms ι and ϕ satisfying the properties above.

4 Further Remarks

1. The categorical picture makes the previously informal phrase “adding a time-axis and then evaluating” mathematically precise.
2. The discrete evaluation operator Eval is the analogue of a Banach contraction step in this setting; it guarantees convergence to the unique 0-dimensional fixed point $\mathbf{o}$.
3. One may enrich the framework by specifying concrete choices for K (e.g. literal string length) and for the set of time tokens $\mathcal{T}$ (e.g. a singleton $\{*\}$), which turns the abstract construction into an implementable algorithm.